MONOTONICITY, POSITIVITY, EDGE DIRECTION AND FACET CONTAINMENT FOR THE FOLDING MAP OF THE ASSOCIAHEDRON

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned publisher source sources/jep-2021-142-associahedra-diagonal/full-original-native.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the retained preamble is the source's own retained-preamble.tex (the class-compatibility file prepared for this source, with \smaller declared by \providecommand) plus the author's macro block at lines 39-303 of sources/jep-2021-142-associahedra-diagonal/full-original-native.tex (everything from the author's first package to the line before \begin{document}, comments excluded; the publisher class, its private \makeatletter block and the conditional font-shape rule are left out deliberately). Nothing was added, no macro was invented, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Assembly history: the first attempt (r001) used inherited-layout-adapter.tex, whose \newcommand{\smaller}{\small} clashes with the amsart wrapper (LaTeX error: command \smaller already defined); r002 switches to the source's retained-preamble.tex, which is the same adapter with \providecommand, and to the author block starting at the first package line. No mathematical text changed between the revisions.

Editorial note. External references are navigation pointers only; every deliberately kept step of the author's argument stays inside the counted spans. The displayed numbers of those pointers are the author's own numbering under the author's \newtheorem declarations (section.index as declared), computed mechanically by the lane's scanner.

Editorial note. Clause (2) of the statement is dismissed by the author with 'This is straightforward from the definition of v' (proof line 1432); the other three clauses carry their own arguments inside the counted span. The sentence is kept verbatim.

Editorial note. The proof reads the clauses in the statement's own order and needs no reference to any item outside the counted range, so no navigation stub was required.

Allowed standard prerequisites (editorial).

- The folding map L from the vertices of the polytope P onto the Boolean lattice, and the vector v defined from the sequence a_i. Set up immediately before the counted statement, in the paragraph ending at source line 1412. Supplies the objects L, v and the increasing sequence a_i in terms of which all four clauses are phrased.
- The adjacencies of the associahedron are generated by rotating pairs of successive left- and right-leaning edges. Used inside the counted proof at lines 1430 and 1434-1438. Lets the author compare two neighbouring trees s and t and compute the edge vector between their images.

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Lemma 1. *The following properties are satisfied:*

(1) $s \leq t \Rightarrow L(s) \leq L(t)$,
(2) $\langle \vec{e}_i, \vec{v} \rangle > 0$, *for* $1 \leq i \leq d$,
(3) *each edge connecting* $s \in L_i^{-1}(0)$ *and* $t \in L_i^{-1}(1)$ *is parallel to* $\vec{e}_i$,
(4) *any fiber* $L^{-1}(b)$, *for* $b \in \{0,1\}^d$, *is contained in a facet of* P.

Proof. (1) When we switch a pair of successive left and right-leaning edges to a pair of successive right and left-leaning edges, either it does not change the orientation of the leaves or it just changes one left-leaning leaf into a right-leaning leaf.
(2) This is straightforward from the definition of $\vec{v}$.
(3) It is enough to prove it on one polytope K_ω. In this case, t is a planar binary tree obtained from a planar binary tree s by switching the $(i+1)^{\text{th}}$ leaf from left-leaning to right-leaning, which implies

$$\overrightarrow{M(s,\omega)M(t,\omega)} = \omega_i \omega_{i+2} \vec{e}_i.$$

(4) Reading the sequence $(1, b_1, \ldots, b_{n_1-2}, 0)$ from left to right, there is at least one occurrence of $(1,0)$, say at place i and $i+1$. Every face labeled by a forest of trees t satisfying $L(t) = b$ lies in the facet labeled by $\left(c_{n_1-1} \circ_i c_2, c_{n_2}, \ldots, c_{n_k}\right)$. □

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